Dynamic Production Planning Under Seasonal and Stochastic Demand:

A Hierarchical Stochastic Decision Support System for B2B Bubble Tea Manufacturing

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Abstract

We develop a three-layer hierarchical decision support system for a B2B bubble tea manufacturer facing seasonal demand uncertainty. The system integrates a monthly two-stage stochastic aggregate model with a daily MILP operational scheduling model via inventory-target linking constraints, executed weekly in a rolling horizon. Exact solvers exceeded 2,200 seconds on full-year daily instances, motivating hierarchical decomposition. Across 100 VSS iterations, the stochastic approach strictly dominated the deterministic approximation. Compared with the current semi-manual system, the proposed approach reduces average total cost by 42% and average overtime from 28.2 to 0.3 days, confirmed through a four-week live pilot.

1 Company Information

Nesco Gıda İçecek (Ankara, Turkey), operating under the brand BobaCo, was among the first manufacturers to produce bubble tea on an industrial scale in Turkey [Nesco Gıda Sanayi ve Ticaret A.Ş., n.d.]. All products are sold exclusively under a B2B model to HORECA customers (cafés, restaurants, hotels, and beverage chains) and exported to over 30 countries. Certifications include VEGAN, ISO 9001, and FDA.

The production portfolio consists of four items across two dedicated lines. Line 1 produces Plastic Tub Boba (a finished product sold directly) and Semi-Finished Boba in Tubs (an intermediate input consumed by Line 2). Line 2 produces Cup Bubble Tea and Can Bubble Tea; due to pasteurization constraints, Cup and Can cannot be produced on the same day, and each line processes at most one product type per day. Products have a shelf life of 12–18 months. The previous facility had negligible storage capacity, making inventory build-ahead infeasible. The new facility provides adequate storage, creating both the opportunity and the operational necessity for a formal production planning system that exploits this shelf life to prepare for the high-demand summer season while managing raw-material lead times of up to 75 days.

2 System Analysis and Project Definition

Figure 1 illustrates the production process flow. The core planning challenge is to balance holding and backlog costs across four products and two production lines, building inventory ahead of the summer peak while managing raw material lead times of up to 75 days and satisfying capacity and line-exclusivity constraints within the new facility's storage. Demand, particularly in the domestic market, fluctuates seasonally and cannot be predicted precisely, so a mathematically based decision support system is required to minimize holding and backlog costs over the planning horizon.

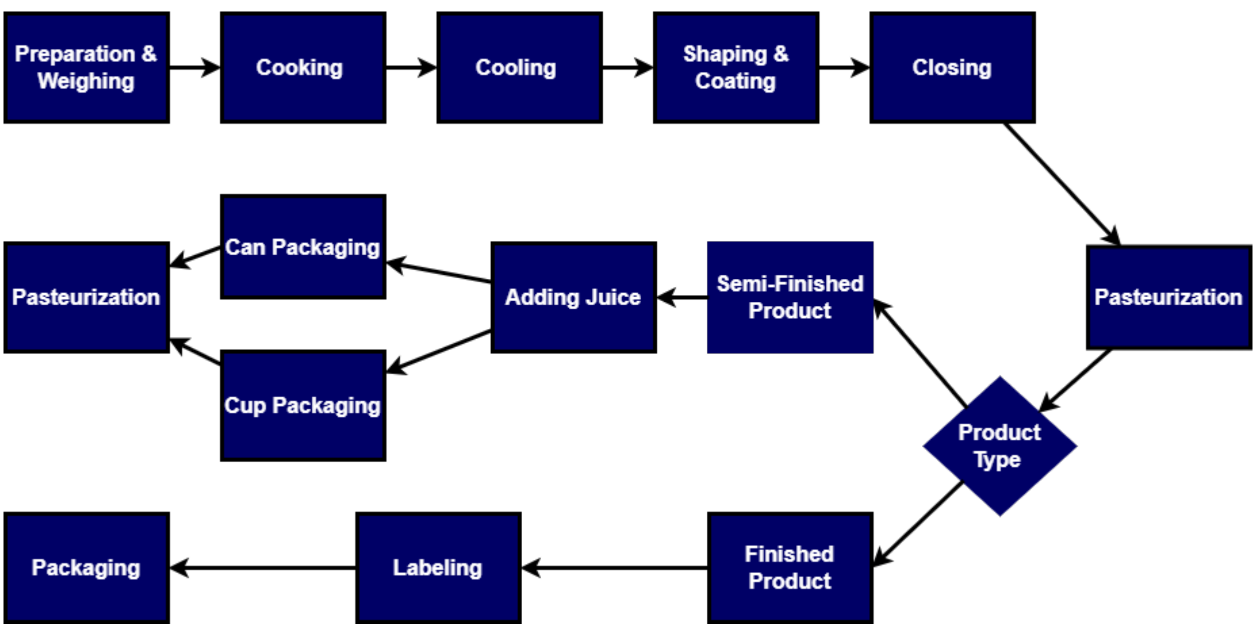

Figure 1: Production workflow of popping boba in the facility

3 Literature Review

Production planning under demand uncertainty has been the subject of sustained research for more than four decades, spanning aggregate planning, hierarchical decomposition, capacitated lot-sizing, scheduling, and stochastic programming. Nahmias and Olsen [2015] present the textbook treatment of the deterministic foundation, including the trade-offs between setup, production, inventory, and backlog. The most cited modern survey of stochastic production planning is Mula et al. [2006], which catalogs over a hundred models that incorporate demand, lead time, or capacity uncertainty and identifies aggregate production planning under uncertainty as the dominant research stream. More recent work by Thevenin et al. [2021] extends this tradition to multi-echelon material requirements planning (MRP) with stochastic optimization, demonstrating that jointly optimizing lot sizes and safety stocks yields substantial cost reductions over decoupled MRP heuristics. These frameworks collectively establish the cost structure, decision typology, and methodological vocabulary that our models inherit.

At the aggregate planning level, BobaCo's long-term model belongs to the family of multi-level capacitated lot-sizing problems (MLCLSP), formally introduced by Billington et al. [1983]. The single-level capacitated lot-sizing problem (CLSP) and its variants are reviewed by Karimi et al. [2003], who emphasize the NP-hardness of even moderately sized instances. The multi-level extension, where intermediate items feed downstream products through a bill-of-materials, is analyzed by Maes et al. [1991], who prove that feasibility of an MLCLSP instance is itself NP-complete. The state-of-the-art exact solution approach is the fix-and-optimize heuristic of Helber and Sahling [2010], which targets medium-scale instances; in our testing, a monolithic full-year daily MILP exceeded 2,200 seconds of solver time, confirming that decomposition is essential at realistic scale.

At the operational level, our daily MILP extends beyond classical MLCLSP. Because production quantities are fully determined by binary machine-assignment variables and because Cup and Can Bubble Tea cannot share a production day, the model integrates lot-sizing decisions with daily machine-assignment scheduling. This places the operational layer in the lot-sizing and scheduling literature, surveyed by Drexl and Kimms [1997], which distinguishes pure lot-sizing problems (quantities only) from combined lot-sizing and scheduling problems (quantities, sequence, and machine assignment). The two layers of our system therefore belong to two distinct but adjacent problem classes, a distinction we exploit in the hierarchical decomposition developed in Section 4.

The closest published structural analogs to our operational layer come from the soft drink industry. Ferreira et al. [2009] formulate a mixed-integer model for a real Brazilian soft drink plant in which liquid preparation in tanks must be synchronized with parallel bottling lines, an arrangement structurally similar to our Semi-Finished Boba feeding the Cup and Can lines. Toledo et al. [2009] address the same plant with a multi-population genetic algorithm and frame it as a synchronized and integrated two-level lot-sizing and scheduling problem. Türkmen et al. [2021] apply multi-stage production planning at a small food and beverage company in Türkiye. Our contribution extends these by integrating a stochastic aggregate layer with a deterministic daily scheduling layer and deploying both in a live rolling-horizon decision support system, a combination not previously published in this domain.

A standard extension of lot-sizing models is to permit backlogging, where unmet demand is carried forward and fulfilled in a later period. The strong reformulations of Pochet and Wolsey [1988] provide the canonical reference for incorporating backlog variables and penalties into MIP lot-sizing models. We extend this framework with a two-tier penalty distinguishing regular backlog from a critical backlog beyond twelve days, the empirically observed threshold above which BobaCo's customers cancel orders. This two-tier formulation is an original modeling contribution motivated directly by industrial data.

Hierarchical production planning (HPP) addresses the complexity of multi-period, multi-product systems by decomposing decisions across temporal aggregations. Bitran et al. [1982] formalize HPP as a two-stage system in which an upper-level aggregate model determines medium-term production targets that a lower-level model disaggregates into detailed schedules. Liberatore and Miller [1985] describe an HPP system deployed at American Olean Tile, where an annual aggregate plan is linked to short-term mixed-integer scheduling and embedded in routine company practice. We follow this two-stage structure, passing monthly inventory targets from the stochastic aggregate model to the short-term scheduling MILP as hard linking constraints at days 24 and 48.

The mathematical mechanism by which one planning level constrains another has been formalized in the capacitated lot-sizing literature. Suerie and Stadtler [2003] introduce an MIP formulation in which setup and inventory variables are explicitly linked across consecutive planning periods, demonstrating that valid inequalities and decomposition heuristics can exploit this structure to achieve tight bounds. The cross-level inventory-target linking constraints we introduce at days 24 and 48 are a direct application of this principle, ensuring that the operational scheduling MILP respects the build-ahead targets identified by the stochastic aggregate model.

Rolling-horizon execution complements hierarchical decomposition by enabling plans to be updated as new demand information becomes available. Sahin et al. [2013] review rolling-horizon schemes in supply chains and conclude that they are effective for maintaining responsiveness in multi-period planning, provided near-term decisions are stabilized to prevent excessive fluctuations. The stability problem itself is studied as schedule nervousness; Blackburn et al. [1986] compare five nervousness-dampening strategies and prove that freezing a portion of the planning horizon dominates safety-stock and lot-for-lot approaches under most cost conditions. Forel and Grunow [2023] extend this stream by combining stochastic lot-sizing with rolling-horizon execution, anticipating forecast updates within the stochastic formulation itself and reporting average cost reductions of fourteen percent over deterministic planning. Near-term decisions in our system are stabilized following the schedule-freezing approach of Blackburn et al. [1986], with weekly replanning aligned to the company's planning cycle.

The stochastic aggregate layer is formulated as a two-stage stochastic program with recourse, following the textbook framework of Birge and Louveaux [2011]. First-stage decisions for the first two months are made before demand uncertainty is revealed; scenario-dependent recourse decisions for months three through twelve allow production, inventory, and backlog quantities to adapt to the realized scenario. Körpeoğlu et al. [2011] apply a multi-stage stochastic programming approach to master production scheduling under demand and capacity uncertainty, demonstrating the value of stochastic formulations for industrial-scale planning. We follow the two-stage formulation in the aggregate layer, making first-stage decisions before uncertainty is revealed and optimizing scenario-dependent recourse over the remaining horizon.

The benefit of solving the stochastic formulation rather than its deterministic mean-value approximation is quantified by the Value of the Stochastic Solution (VSS), formally introduced by Birge [1982] for stochastic linear programs with fixed recourse. Thevenin et al. [2021] compute VSS in a production planning context and confirm that the stochastic approach yields cost savings that grow with demand volatility and capacity tightness. Our 100-iteration VSS analysis applies this definition; the stochastic approach dominated the deterministic approximation in every iteration, with the stochastic solution achieving on average only 10.7% of the cost of the expected-value solution under uncertainty.

Taken together, these literature streams motivate the hierarchical stochastic rolling-horizon architecture developed in Section 4. The aggregate planning layer is a two-stage stochastic MLCLSP that determines monthly production and inventory targets. The operational layer is a multi-level lot-sizing and scheduling MILP that solves daily machine-assignment decisions subject to the aggregate-layer linking constraints. Both layers are re-solved weekly in a rolling horizon with near-term stabilization. The closest published analog, Englberger et al. [2016], integrates two-stage stochastic master production scheduling with rolling-horizon execution but does not separate the operational scheduling decisions into a distinct MILP layer with hard cross-level linking constraints. To the best of our knowledge, no prior study reports a deployed industrial decision support system that combines all three architectural elements — a two-stage stochastic aggregate model, a separate daily operational scheduling MILP linked by explicit inventory targets, and weekly rolling-horizon execution with near-term stabilization — in a single hierarchical stack.

4 Proposed Solution Strategy

4.1 Problem Classification and Architectural Overview

The planning problem faced by BobaCo spans two interconnected problem classes. At the aggregate planning level, the problem belongs to the multi-level capacitated lot-sizing problem (MLCLSP) family under demand uncertainty [Billington et al., 1983, Karimi et al., 2003, Maes et al., 1991, Körpeoğlu et al., 2011], where decisions concern monthly production volumes and inventory targets for a multi-level bill-of-materials with one intermediate product (Semi-Finished Boba) that is both sold directly and consumed by two finished products. At the operational scheduling level, the problem integrates lot-sizing with daily machine-assignment decisions, including a product-exclusivity constraint between Cup and Can Bubble Tea on the pasteurization line; this structure is classified by Drexl and Kimms [1997] as a multi-level lot-sizing and scheduling problem. Even state-of-the-art exact methods such as the fix-and-optimize approach of Helber and Sahling [2010] target medium-scale instances; in our testing, a monolithic full-year daily MILP exceeded 2,200 seconds, confirming that the combined formulation is computationally intractable at planning resolution and motivating the hierarchical decomposition developed below.

To the best of our knowledge, this is the first study to design, mathematically formulate, and deploy a live three-layer hierarchical decision support system that integrates a monthly two-stage stochastic aggregate planning model with a daily MILP operational scheduling model via temporal inventory-target linking constraints under a weekly rolling-horizon execution in a B2B process manufacturing setting. Existing studies address stochastic rolling-horizon planning [Forel and Grunow, 2023], integrated lot-sizing and scheduling under uncertainty [Drexl and Kimms, 1997, Ferreira et al., 2009, Toledo et al., 2009], or stochastic multi-echelon MRP [Thevenin et al., 2021], but no published live industrial system combines all three architectural elements in one deployed stack [Englberger et al., 2016].

Contributions. This paper makes three contributions:

1. **Architecture.** We formulate the first three-layer hierarchical stochastic rolling-horizon decision support system for B2B process manufacturing, combining a two-stage stochastic aggregate model, a daily MILP operational scheduling model with machine-assignment binaries, and weekly rolling-horizon execution with near-term stabilization. Computational experiments show that the monolithic daily MILP is intractable at full planning resolution (Scenario 9 exceeded 2,200 seconds), establishing that hierarchical decomposition is computationally necessary.

2. **Modeling.** We introduce a two-tier backlog penalty that separately tracks demand unmet beyond 12 days — the empirically observed order-cancellation threshold — extending the canonical backlogging formulation of Pochet and Wolsey [1988] with an industrially motivated distinction between operationally tolerable delays and service failures with direct revenue consequences.

3. **Deployment.** We deploy the system in a four-week live pilot integrated into the company's actual weekly planning cycle. Benchmarked against the existing semi-manual approach across 10 demand scenarios, the system reduces average total cost by 42%, cuts average overtime from 28.2 to 0.3 days, and achieves zero backlog and zero overtime in 7 of 10 scenarios.

Across 100 VSS iterations, the stochastic formulation strictly dominates the deterministic approximation.

4.2 Modeling Approach and Key Assumptions

The system is modeled as a MILP over a daily planning horizon. Backlogging is permitted to preserve feasibility during temporary shortages, but backlogs exceeding 12 days trigger the critical-backlog penalty b_i^{12} because delays beyond that threshold risk order cancellations. Each production line processes at most one product type per day; Semi-Finished Boba and Plastic Tub Boba are treated as distinct products despite sharing Line 1. Because the monolithic full-year daily MILP is computationally intractable (Scenario 9 exceeded 2,200 seconds), the solution decomposes into a long-term stochastic aggregate model and a short-term rolling-horizon MILP, linked by inventory-target constraints at days 24 and 48.

4.3 Mathematical Model

4.3.1 Formulation of the Mathematical Model

Table 1: Sets of Mathematical Model

Sets	**Definition**
K_1	Set of Plastic Tub and Semi-Finished Boba Products ($K_1 = \{1, 2\}$)
K_2	Set of Cup Bubble Tea and Can Bubble Tea Products ($K_2 = \{3, 4\}$)
T	Set of Days in Period
S	Set of Raw Materials
M	Set of machines on Line 1 ($M = \{1, 2, 3\}$)
M_3	Set of machines for product 3 on Line 2 ($M_3 = \{1, 2\}$)

Table 2: Parameters of Mathematical Model

Parameters	**Definition**
b_i	Daily unit backlog cost of product i
b_i^{12}	Daily unit backlog cost of product i whose due date has exceeded at least 12 days
n_{is}	Number of raw material s needed for one unit of product i
LT_s	Lead time of raw material s
κ_m	Daily capacity of machine $m \in M$ on Line 1 (full capacity if selected)
κ_m^3	Daily capacity of machine $m \in M_3$ when producing product 3 on Line 2
κ^4	Daily capacity of Line 2 when producing product 4 (single machine)
u_{23}	Amount of product 2 needed to produce product 3
u_{24}	Amount of product 2 needed to produce product 4
$d_{i,t}$	Customer demand of product i at day t
h_i	Unit holding cost of product i
h_s^r	Unit holding cost of raw material s (used with weekly RM inventory)
I^{max}	Maximum inventory area
v_i	Inventory area usage of unit product i
I_r^{max}	Maximum raw material inventory area
v_s	Inventory area usage of unit raw material s
$w_{i,t}$	Sum of the demand obtained for product i from period $t-12$ to t (Demand of last 12 days)
moq_s	Minimum order quantity for raw material s
$I_{i,0}$	Initial inventory of product i at the start of day 1
$B_{i,0}$	Initial backlog of product i at the start of day 1
$I_{s,0}^r$	Initial raw material inventory of s at the start of day 1

$$\min \sum_{t\in T}\sum_{i\in K_1\cup K_2}\left(b_i B_{i,t} + b_i^{12} B_{i,t}^{12} + h_i I_{i,t}\right) + \sum_{t\in T}\sum_{s\in S} h_s^r I_{s,t}^r \tag{1}$$

$$\text{s.t.} \quad X_{m,1,t} + X_{m,2,t} \leq 1 \qquad \forall m \in M, \forall t \in T \tag{2}$$

$$X_{m,3,t} \leq 1 - X_{4,t} \qquad \forall m \in M_3, \forall t \in T \tag{3}$$

$$P_{1,t} = \sum_{m\in M} \kappa_m X_{m,1,t} \qquad \forall t \in T \tag{4}$$

$$P_{2,t} = \sum_{m\in M} \kappa_m X_{m,2,t} \qquad \forall t \in T \tag{5}$$

$$P_{3,t} = \sum_{m\in M_3} \kappa_m^3 X_{m,3,t} \qquad \forall t \in T \tag{6}$$

$$P_{4,t} = \kappa^4 X_{4,t} \qquad \forall t \in T \tag{7}$$

$$I_{2,t-1} + P_{2,(t-1)} - \sum u_{2i} P_{i,t} = I_{2,t} \qquad \forall t \in T \tag{8}$$

Table 3: Decision Variables of Mathematical Model

Variables	Definition
$P_{i,t}$	Number of unit i produced at day t
$I_{i,t}$	Number of unit product i in inventory at day t
$I^r_{s,t}$	Number of raw material s in inventory at the end of day t
$R_{s,t}$	Number of raw material s ordered at day t
$Z_{s,t}$	1 if an order is placed for raw material s at day t, 0 otherwise
$B_{i,t}$	Backlog amount of product i at day t
$B^{12}_{i,t}$	Backlog amount of product i at day t whose due date has exceeded at least 12 days
$X_{m,1,t}$	1 if machine $m \in M$ produces product 1 at day t, 0 otherwise
$X_{m,2,t}$	1 if machine $m \in M$ produces product 2 at day t, 0 otherwise
$X_{m,3,t}$	1 if machine $m \in M_3$ produces product 3 at day t, 0 otherwise
$X_{4,t}$	1 if Line 2 produces product 4 at day t, 0 otherwise

The two-tier backlog penalty ($B_{i,t}, B^{12}_{i,t}$) extends the standard backlogging formulation of Pochet and Wolsey [1988] by separately tracking demand unmet for more than 12 days — the empirically observed threshold beyond which BobaCo's customers cancel orders. This two-tier structure is an original modeling contribution motivated by industrial data; it allows the model to distinguish operationally tolerable short delays from service failures with direct revenue consequences, and is reflected in the differentiated unit cost coefficients b_i and b^{12}_i in the objective function.

For the deployed instance, the short-term MILP over the 48-day rolling window has 432 binary variables (288 on Line 1, 96 on Line 2 product 3, and 48 on Line 2 product 4) plus raw material order binaries, and approximately 770 continuous variables for production, inventory, backlog, and critical-backlog flows. The full-year monolithic exact formulation ($|T| = 288$) thus exceeds 2,500 binaries and 4,600 continuous variables — explaining the empirically observed intractability of the monolithic solve.

4.4 Heuristic Approach

In this heuristic approach, we divide the original model into two linked models: a long-term model and a short-term model. The short-term model is essentially the same mathematical model as the original one, but it is solved only for a 48-day horizon, which corresponds to the number of working days in 8 weeks. In this model, we still decide the daily production and inventory levels for each product on each line, and we add two extra parameters to incorporate long-term inventory decisions. These additional parameters come from the long-term model.

The long-term model works with monthly time periods over the whole year and shows how much we plan to produce and keep in inventory at an aggregate level. To account for demand uncertainty in this planning layer, the long-term model is formulated as a two-stage stochastic program. The first stage covers the initial months (t = 1,2), where decisions are made before uncertainty is revealed. The second stage covers the remaining months (t = 3,...,12), where demand is represented through scenarios $\omega \in \Omega$ and scenario-dependent recourse decisions are optimized. The objective combines the first-stage costs with the expected second-stage costs across scenarios, and the system state at the end of t = 2 is carried into the scenario-based stage to ensure consistency.

The inventory decisions obtained from this long-term model are then used as hard linking constraints in the short-term model, ensuring that the operational scheduling decisions respect the build-ahead targets identified by the stochastic aggregate model — a coordination mechanism formalized in the capacitated lot-sizing literature by Suerie and Stadtler [2003]. Both models are run in a rolling-horizon manner [Sahin et al., 2013, Forel and Grunow, 2023]: the long-term model is solved monthly with the necessary parameter updates, and the short-term model is repeatedly solved weekly over a 2-month time period (48 workdays). Near-term decisions are stabilized following the schedule-freezing approach of Blackburn et al. [1986], which has been shown to dominate alternative nervousness-dampening strategies in rolling-horizon production planning. The combination of the two models provides a detailed short-term production schedule while still considering future demand and inventory costs. Figure 2 demonstrates a chart to see the two models logic visually:

4.4.1 Deterministic Long-Term Aggregate Production Planning Model

This long-term model covers similar constraints to the mathematical model, with the exception of strict daily production capacity. Moreover, this model's constraints cover monthly decisions and do not consider raw material existence. The objective of the long-term model is to find a less costly monthly long-term inventory strategy by minimizing holding and backlog costs to cover the high-demand season with an optimal inventory level.

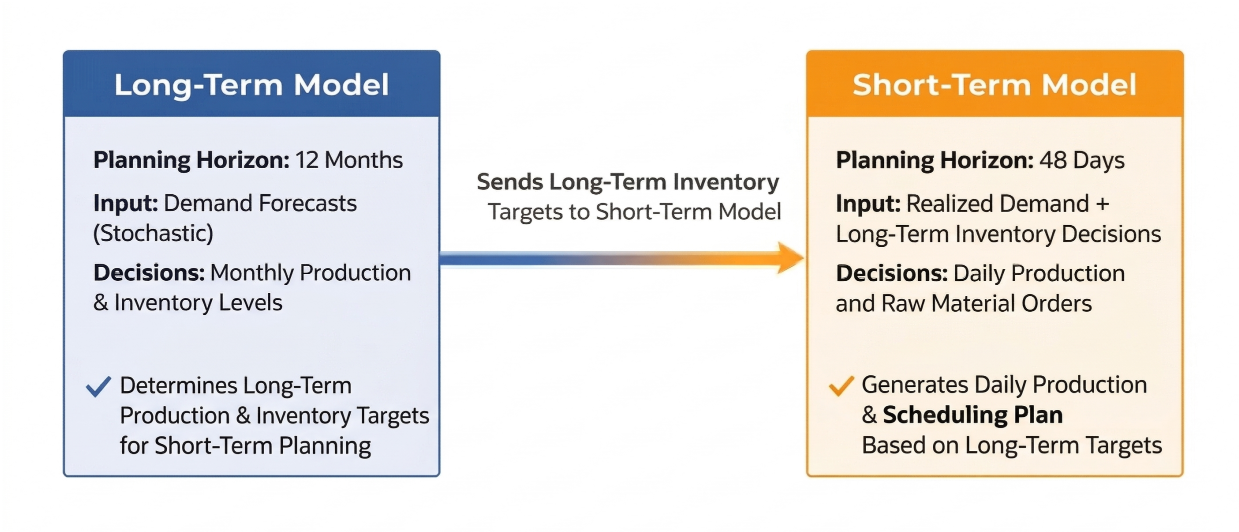

Figure 2: Three-layer hierarchical architecture: a monthly two-stage stochastic aggregate model passes inventory targets to a daily operational MILP via linking constraints at days 24 and 48, both re-solved weekly in a rolling horizon with near-term stabilization.

Table 4: Sets of Deterministic Long-Term Model

Sets	Definition
K_1	Set of Plastic Tub and Semi-Finished Boba Products
K_2	Set of Cup Bubble Tea and Can Bubble Tea Products
T	Set of Planning Periods ($T = \{1, \ldots, 12\}$)
J	Set of Production Lines

Table 5: Parameters of Deterministic Long-Term Model

Parameters	Definition
$d_{i,t}$	Demand of product i in period $t \in \{1, 2\}$
a_t	Number of workdays in period t
b_i	Backlog cost for product i
c_1	Daily unit capacity of production line 1
$c_{2,i}$	Daily unit capacity of production line 2 for product i
I^{max}	Maximum storage area
r_i	Unit usage area of product i
u_{i2}	Number of product 2 used in the production of product $i \in \{3, 4\}$
h_i	Holding cost of product i for unit period

$$\min \sum_{t=1}^{2} \sum_{i \in K_1 \cup K_2} (b_i B_{i,t} + h_i I_{i,t}) \tag{1}$$

$$\text{s.t.} \quad \sum_{i \in K_1} \frac{P_{i,t}}{c_1} \leq a_t \qquad \forall t \in \{1, 2\} \tag{2}$$

$$\sum_{i \in K_2} \frac{P_{i,t}}{c_{2,i}} \leq a_t \qquad \forall t \in \{1, 2\} \tag{3}$$

Table 6: Decision Variables of Deterministic Long-Term Model

Variables	**Definition**
$P_{i,t}$	Production amount of product i at period $t \in \{1, 2\}$
$I_{i,t}$	Inventory amount of product i at the end of period $t \in \{1, 2\}$
$B_{i,t}$	Backlog amount of product i at period $t \in \{1, 2\}$

4.4.2 Two-Stage Stochastic Long-Term Aggregate Production Planning Model

To include demand uncertainty in the solution approach, the long-term model is formulated as a two-stage stochastic program with recourse [Birge and Louveaux, 2011], where first-stage production and inventory decisions for the early months are fixed before uncertainty is revealed and second-stage recourse decisions adapt to the realized demand scenario in the remaining months. Multi-stage stochastic programming applications to master production scheduling under demand uncertainty are surveyed by Körpeoğlu et al. [2011] and Thevenin et al. [2021].

The stochastic model extends the notation of the deterministic model above with: Ω (set of demand scenarios), $p_\omega = 1/|\Omega|$ (equal scenario probability), $d_{i,t,\omega}$ (demand of product i in period $t \in \{3, \ldots, 12\}$ under scenario ω), and scenario-indexed recourse variables $P_{i,t,\omega}$, $I_{i,t,\omega}$, $B_{i,t,\omega}$ for $\omega \in \Omega$.

$$\min \sum_{t=1}^{2} \sum_{K_1 \cup K_2} \big(b_i B_{i,t} + h_i I_{i,t}\big) + \sum_{\omega \in \Omega} \sum_{t=3}^{12} \sum_{i \in K_1 \cup K_2} p_\omega \big(b_i B_{i,t,\omega} + h_i I_{i,t,\omega}\big) \tag{1}$$

$$\text{s.t.} \quad \sum_{i \in K_1} \frac{P_{i,t}}{c_1} \leq a_t \qquad \forall t \in \{1,2\} \tag{2}$$

$$\sum_{i \in K_2} \frac{P_{i,t}}{c_{2,i}} \leq a_t \qquad \forall t \in \{1,2\} \tag{3}$$

$$\sum_{i \in K_1} \frac{P_{i,t,\omega}}{c_1} \leq a_t \qquad \forall t \in \{3,12\}, \forall \omega \in \Omega \tag{4}$$

$$\sum_{i \in K_2} \frac{P_{i,t,\omega}}{c_{2,i}} \leq a_t \qquad \forall t \in \{3,12\}, \forall \omega \in \Omega \tag{5}$$

$$I_{2,t-1} + P_{2,t} - \sum_{i \in K_2} u_{i2}\, P_{i,t} = I_{2,t} \qquad \forall t \in \{1,2\} \tag{6}$$

$$I_{2,2} + P_{2,3,\omega} - \sum_{i \in K_2} u_{i2}\, P_{i,3,\omega} = I_{2,3,\omega} \qquad \forall \omega \in \Omega \tag{7}$$

$$I_{2,t-1,\omega} + P_{2,t,\omega} - \sum_{i \in K_2} u_{i2}\, P_{i,t,\omega} = I_{2,t,\omega} \quad \forall t \in \{4,\ldots,12\} \tag{8}$$

$$I_{i,t-1} + P_{i,t} - (d_{i,t} + B_{i,t-1}) + B_{i,t} = I_{i,t} \tag{9}$$

$$\forall t \in \{1,2\}, \forall i \in (K_1 \cup K_2) \backslash \{2\} \tag{9}$$

$$I_{i,2} + P_{i,3,\omega} - (d_{i,3,\omega} + B_{i,2}) + B_{i,3,\omega} = I_{i,3,\omega} \tag{10}$$

$$\forall \omega \in \Omega, \forall i \in (K_1 \cup K_2) \backslash \{2\} \tag{10}$$

$$I_{i,t-1,\omega} + P_{i,t,\omega} - (d_{i,t,\omega} + B_{i,t-1,\omega}) + B_{i,t,\omega} = I_{i,t,\omega} \tag{11}$$

$$\forall t \in \{4,\ldots,12\}, \forall \omega \in \Omega, \forall i \in (K_1 \cup K_2) \backslash \{2\} \tag{11}$$

$$I^{max} \geq \sum_{i=1}^{4} r_i I_{i,t} \qquad \forall t \in \{1,2\} \tag{12}$$

$$I^{max} \geq \sum_{i=1}^{4} r_i I_{i,t,\omega} \qquad \forall t \in \{3,\ldots,12\}, \forall \omega \in \Omega \tag{13}$$

$$P_{i,t}, I_{i,t}, B_{i,t}, P_{i,t,\omega}, I_{i,t,\omega}, B_{i,t,\omega} \geq 0 \tag{14}$$

$$\forall i \in K_1 \cup K_2, \forall t \in T, \forall \omega \in \Omega \tag{14}$$

4.4.3 Scenario Generation for the Two-Stage Long-Term Model

Scenarios are generated from company-provided monthly forecasts f_t and deviation rates s_t for months 3–12. Each scenario k draws independently from Uniform$[f_t(1 - s_t),\ f_t(1 + s_t)]$; months 1–2 are treated as deterministic first-stage inputs. Equal probabilities $p_k = 1/N$ are assigned when no additional information is available.

4.4.4 Short-Term Production Planning Model

The short-term model uses the same sets (K_1, K_2, T, S, M, M_3), parameters, and decision variables as the Mathematical Model (Section 4.3.1), including the machine-assignment binary variables $X_{m,j,t}$ and $X_{4,t}$. Two additional linking parameters are introduced:

$d^D_{i,1}$	Minimum inventory level of product i required at day 24, as specified by the long-term model output
$d^D_{i,2}$	Minimum inventory level of product i required at day 48, as specified by the long-term model output

The short-term model applies the same objective function and constraints (1)–(14) and (17)–(20) as the Mathematical Model (Section 4.3.1), restricted to a 48-workday rolling window ($|T| = 48$). The heuristic adds two inventory-target constraints that enforce the aggregate-layer build-ahead targets at the linking days:

$$I_{i,24} \geq d^D_{i,1} \quad \forall i \in K_1 \cup K_2 \tag{15}$$

$$I_{i,48} \geq d^D_{i,2} \quad \forall i \in K_1 \cup K_2 \tag{16}$$

Constraints (15) and (16) implement the cross-level inventory coordination mechanism of Suerie and Stadtler [2003]: the operational MILP cannot produce a schedule that falls below the aggregate-layer targets at days 24 and 48, ensuring that the build-ahead strategy identified by the long-term model is realized in the daily schedule.

4.5 Verification

4.5.1 Verification of the Mathematical Model

The mathematical model was verified along two axes: internal logical consistency under controlled test cases, and computational scalability across nine demand scenarios with increasing seasonality. Nine structured test cases confirmed that cost trade-offs, raw material dependencies, and feasibility behave consistently with theoretical expectations (Table 7).

Table 7: Logical verification test cases and model responses

Case	Operating Condition	Observed Response
I	Equal backlog and holding costs	Production clustered near demand period
II	Backlog cost > holding cost	Production advanced; monotone in cost ratio
III	Holding cost > backlog cost	Production delayed; inventory near zero
IV	No initial raw materials	Production deferred to first material arrival
V	High raw material holding cost	Materials consumed immediately upon arrival
VI	Single-product demand	Consistent plans at all demand magnitudes
VII	Universally high demand	Feasibility maintained under extreme load
VIII	Universally low demand	Zero-backlog plan with minimal inventory
IX	Near-zero lead times	Markedly elevated computation time

For scalability, nine demand scenarios with monotonically increasing seasonality were solved: Scenarios 1–7 completed rapidly, Scenario 8 showed pronounced runtime growth, and Scenario 9 exceeded 2,200 seconds. This runtime spike empirically confirms that the monolithic exact MILP is computationally intractable at full planning resolution, necessitating the hierarchical heuristic.

4.5.2 Verification of the Hierarchical Heuristic

The exact solution and the heuristic were compared on a full-year horizon under identical demand, tracking three indicators: total objective value, backlog exceeding ten days, and average lateness per unit. In the company's typical below-capacity operating range, the heuristic's total objective stayed within 20% of the exact solution while keeping ten-day backlog below 0.1% of annual orders. In above-capacity stress scenarios — which the exact solver cannot complete in tractable time anyway — the heuristic produced approximately 4 percentage points more ten-day backlog than the exact benchmark, driven by timing mismatches under clustered demand. Figure 3 shows that the heuristic reproduces the exact solution's backlog and inventory logic in the operationally relevant regime, supporting its use as a practical substitute given that the exact formulation is computationally intractable at full scale.

Scen.	Method	Demand	Backlog	Backlog 12	Inventory	RawMat.Inv	Objective	Avg. Late	Opt. Gap
1	Gurobi	351300	267	0	1533926	15069500	3040876	0.00076	–
1	Heuristic	351300	267	0	2332351	12250600	3557411	0.00076	%17.0
2	Gurobi	437100	293	0	2615749	13947600	4010509	0.00067	–
2	Heuristic	437100	60262	0	3614643	11806900	4795333	0.13787	%19.6
3	Gurobi	610000	2088	0	9178206	9229800	10101186	0.00342	–
3	Heuristic	610000	70215	0	11152983	8670300	12020013	0.11511	%19.0
4	Gurobi	762000	2650	0	5406554	6240700	6030624	0.00348	–
4	Heuristic	762000	123890	0	6195144	6220900	6817234	0.16259	%13.0
5	Gurobi	778500	1271	0	19648951	5834200	20232371	0.00163	–
5	Heuristic	778500	16038	0	23141118	3838000	23524918	0.02060	%16.3
6	Gurobi	814000	5820	0	65053896	9399300	65993826	0.00715	–
6	Heuristic	814000	24	0	72807450	7983700	73605820	0.00003	%11.5
7	Gurobi	825000	237563	0	4911561	3238600	5235421	0.28796	–
7	Heuristic	825000	772495	148	4298543	3743800	6152923	0.93636	%17.5
8	Gurobi	827000	91964	0	10281598	3522400	10633838	0.11120	–
8	Heuristic	827000	190144	0	12291378	3428800	12634258	0.22992	%18.8
9	Gurobi	1500000	59095577	49111510	13715055	3578900	4.91E+11	39.39700	–
9	Heuristic	1500000	60575376	50381931	12705679	3868900	5.04E+11	40.38400	%2.6
10	Gurobi	1600000	59735825	47625376	2891913	3621800	4.76E+11	37.33500	–
10	Heuristic	1600000	62095787	49435371	2555550	3734500	4.94E+11	38.81000	%3.8

Figure 3: Heuristic vs. exact solver: backlog and inventory trajectories under typical (below-capacity) and stress (above-capacity) demand conditions, demonstrating that the heuristic reproduces the exact solution's qualitative behavior within the company's normal operating range.

4.5.3 Verification of the Stochastic Solution Approach

The stochastic model was verified against the deterministic model: with a single scenario, both produced identical inventory and production decisions. A VSS analysis [Birge, 1982] was then conducted across 100 iterations with 25% demand variance. For this minimization problem,

$$VSS = EEV - RP$$

where RP is the stochastic recourse problem value and EEV is the expected performance of the expected-value solution under uncertainty. The VSS was non-negative in all 100 iterations (Table 8); a minimum of exactly zero arises in the rare degenerate iterations where the expected-value solution happens to be feasible under the realized scenarios, and the stochastic approach strictly improves on the deterministic approximation in the remainder. Cost values are reported in model units derived from the company's relative penalty structure (holding:backlog:critical backlog cost ratio of 1:160:10,000); they should be interpreted as model-based performance measures rather than direct accounting costs.

Table 8: VSS analysis results across 100 iterations (model cost units).

Average RP Objective Value	4,454,400
Average EEV Objective Value	41,626,235
Average VSS ($EEV - RP$)	37,171,834
Average VSS (avg of %)	80.48%
Min VSS	0.00
Max VSS	87,435,901

Under the yearly rolling horizon, the stochastic model produces higher inventory build-ahead than the deterministic model because the risk of high-cost backlog incentivizes proactive stock accumulation. As shown in Table 9, this increases holding cost but substantially reduces backlog cost.

Table 9: Cost comparisons of stochastic and deterministic models (model cost units; cost ratio holding:backlog:critical backlog = 1:160:10,000).

	Stochastic	Deterministic	Difference
Avg. Holding Cost	5,634,045	2,901,458	-2,732,587
Avg. Backlog Cost	48,057,008	89,656,018	41,599,010
Avg. Total Cost	53,691,052	92,557,476	38,866,424

Sensitivity tests varying the number of scenarios (5–20) and variance (5%–25%) produced monotonically consistent results, confirming correct model behavior.

4.6 Validation

Validation used real company input data and forecast structures. The model builds inventory ahead during low-demand months and draws it down during the summer peak — consistent with the company's stated planning objective. When rolling-horizon forecasts decrease, production levels fall accordingly; when they increase, available capacity and prior inventory are mobilized to limit backlog accumulation. The industrial advisor confirmed this adaptive behavior as operationally realistic. Figure 4 shows the yearly production-demand balance, demonstrating feasibility and interpretable planning patterns throughout the horizon.

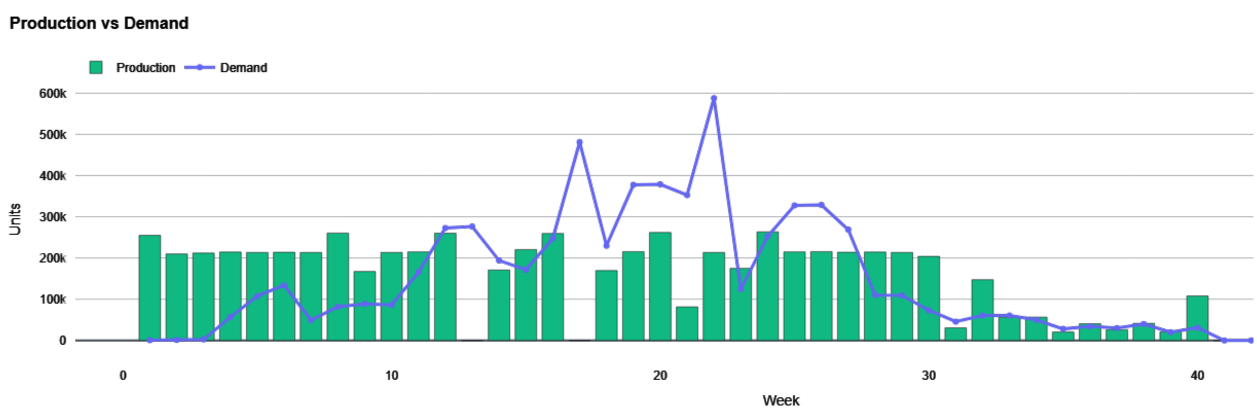

Figure 4: Production and Demand Levels Throughout the Year

5 Deliverables, Benchmarking, and Pilot Deployment

5.1 Deliverables

The primary deliverable is a Python/Streamlit web application (the Weekly Planner) that accepts user-provided forecasts, capacity parameters, and cost coefficients; solves the hierarchical stochastic model; and outputs daily production line assignments, period-by-period raw material orders, and interactive charts. Long-term parameters are revised monthly and short-term data weekly; all tabular results export to Excel. A user manual documents configuration, parameter updates, and output interpretation.

5.2 Benchmarking and Benefits to the Company

After completing the validation phase, a benchmarking analysis was conducted to compare the proposed decision support system with the company's current planning approach under the same demand inputs. The current system was approximated by a 35-day backward allocation rule, calibrated against the company's historical production logs and confirmed by the industrial advisor as a faithful proxy for the semi-manual planning practice in use prior to deployment: production for a given delivery is scheduled to begin 35 days in advance with no proactive build-ahead beyond that window. Since the company reacts to delays exceeding 12 days by using overtime, benchmarking focuses on KPIs related to backlog risk and overtime dependency.

Three KPIs were tracked: backlog exceeding 12 days (cancellation risk), overtime days required, and overtime-dependent delivery rate.

Under the original 2026 forecast, the current system required 11 overtime days and 380,709 units of cancellation-risk backlog (6.01% of demand); the proposed system required none. Across 10 additional scenarios with up to 25% monthly demand deviation, the current system averaged 28.2 overtime days (range 17–37) and a 12.49% risk ratio. The proposed system achieved zero backlog and zero overtime in 7 of 10 scenarios; in the remaining 3 it required only 1 overtime day. Average overtime fell to 0.3 days and residual backlog to 0.08% of demand (Table 10).

Table 10: Benchmarking Results for the Current and Proposed Systems

Metric	**Current System**	**Proposed System**
Base-case overtime days	11	0
Base-case overtime-dependent volume	380,709	0
Base-case overtime delivery rate	6.01%	0.00%
Average overtime days across 10 scenarios	28.2	0.3
Range of overtime days across scenarios	17–37	0–1
Weighted overtime-related / delayed volume ratio	12.49%	0.08%
Scenarios with zero backlog and zero overtime	0/10	7/10

5.3 Implementation and Pilot Study

The system was deployed in a four-week pilot integrated into BobaCo's weekly planning cycle. An interface aligned with the firm's existing reporting practices was developed with company input (Figure 5). The first week ran under project-team supervision; subsequent weeks were operated independently by company planners. Each Monday, the tool generated recommended production quantities and raw material orders, which were reviewed for shop-floor feasibility before execution.

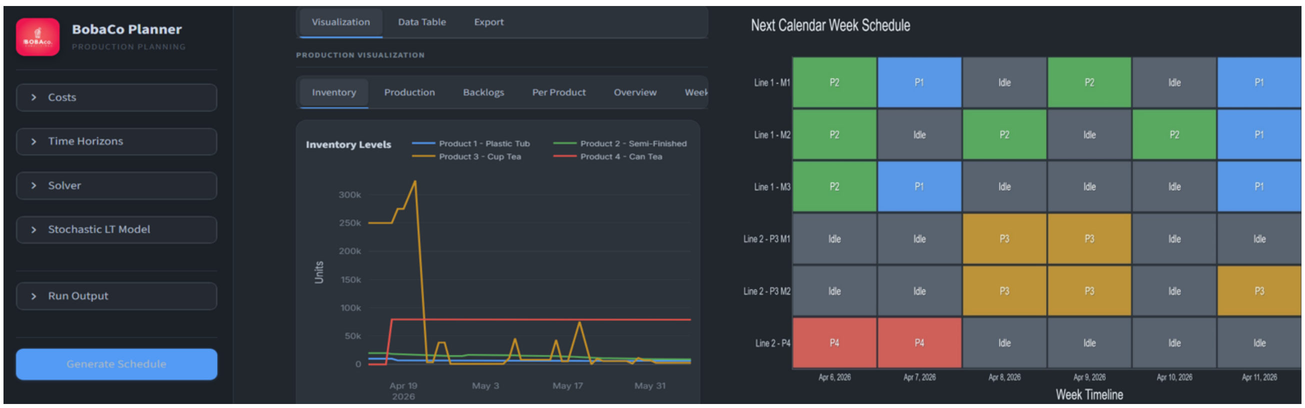

Figure 5: System interface and sample weekly schedule output.

The company approved every weekly plan as feasible. In several weeks the system recommended slightly higher production than the company's baseline, generating inventory buffer ahead of the anticipated summer peak — the primary design objective. The pilot confirmed that the system's weekly decisions are understandable, feasible, and consistent with proactive inventory build-ahead.

6 Conclusion and Future Work

We demonstrate that hierarchical stochastic rolling-horizon decomposition is both computationally necessary and practically effective for B2B seasonal manufacturing. The monolithic daily MILP is intractable at realistic scale (Scenario 9 exceeded 2,200 seconds), motivating the proposed three-layer architecture: a monthly two-stage stochastic aggregate model, a daily MILP operational scheduler with machine-assignment binaries, and weekly rolling-horizon execution with near-term stabilization. Under the yearly rolling horizon, the stochastic approach reduces average total cost by 42.0% relative to the deterministic approximation. The improvement primarily reflects the cost asymmetry inherent to the problem: backlog penalties are substantially higher than holding costs, so the stochastic approach's proactive inventory build-ahead avoids disproportionate backlog losses under uncertainty.

The practical value of the system was also demonstrated through benchmarking and pilot use. Compared with the company's current planning logic, the proposed approach substantially reduced overtime dependency, lowering average overtime requirement from 28.2 days to 0.3 days and achieving zero backlog and zero overtime in 7 out of 10 generated demand scenarios. In addition, the four-week pilot study showed that the system could be incorporated into the company's weekly planning routine and could support inventory positioning for the high demand anticipated in the 2026 summer forecast. The hierarchical stochastic rolling-horizon approach developed here is directly applicable to any SME facing seasonal demand with long-horizon procurement lead times and recently expanded storage capacity, particularly in food and beverage manufacturing where multi-level BOM dependencies and synchronized line constraints are common. Future work may focus on improving the scenario generation structure through observations collected over longer periods, allowing demand uncertainty to be represented in an even more realistic way.

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